Allen Abraham

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Professional Summary

Mid-level software engineer specializing in cloud pipelines (**AWS** and **GCP**), **distributed systems, LLM orchestration**. Leveraging AI-driven development like **Claude Code** to maximize velocity, drive rapid prototyping, and deliver production-ready solutions, like an **Airflow DAG** that cut data ingestion time by 99.8% and a refunds platform handling over 1 million transactions per hour with **99.999% uptime**. Experienced in building scalable **microservices**, ETL workflows, and real-time monitoring dashboards. Looking to leverage my expertise to drive **reliable**, **high-performance** infrastructure in a fast paced environment.

Education

Master of Science in Computer Science | *New York University* — 08/**2023** - 05/**2025**

GPA: **3.96** / 4.0

Bachelor of Engineering in Computer Science | *Visvesvaraya Technological University* — 08/**2017** - 05/**2021**

GPA: **8.33** / 10.0

Work Experience

Canonical (Ubuntu) | *Software Engineer* — 08/**2025 - Present**

- Modernize and migrate the **Jenkins** architecture to **Kubernetes** using **Terraform** and **Juju**, reducing deployment time and increasing system **reliability.**
- Lead the **AI initiative** for the Public Cloud team, integrating tools like **Claude Code** into the workflow to increase developer productivity and experience. Designing an **agentic** tool with **OpenRouter** and **Python** to streamline **Jenkins** pipeline debugging
- **Modernizing** the Ubuntu build tool using imagecraft, leading to performant and maintainable tooling for latest Ubuntu builds.
- Improving **integration tests** of builder images to catch bugs and misses in our pipelines and to produce more stable Ubuntu images

Firebird Music | *Data Engineer Intern* — 06/**2024** - 08/**2024**

- Developed an **Airflow DAG** on Cloud Composer to incrementally load data from an S3 bucket into partitioned **BigQuery** tables via Cloud Storage, reducing ingestion time by **99.8%** and saving **$10K** annually by eliminating full table refreshes.
- Collaborated with a globally distributed team to analyze and interpret social media images for over 50,000 artists using the **GPT-Vision** API, uncovering engagement trends that informed marketing strategy and guided artists on image-driven improvements.

Juspay | *Software Engineer* — 11/**2020** - 07/**2023**

- Optimized gateway scoring service by implementing **Groovy-based algorithms**, increasing customer success rates by **up to 30%.**
- Owned the refunds vertical, introducing features such as a **PostgreSQL** and Redis-based distributed producer–consumer workflow and scalable, **REST APIs** - scaling to process over **1 million refunds** per hour with a median transaction amount of $100.
- Built Prometheus and ClickHouse Grafana dashboards on GCP for **real-time metric** analysis, maintaining 98% refund success & **99.999% uptime**, slashing debugging and incident response times (**hours to minutes**), and aiding cross-functional product architecture improvements.
- Led backend development in **Haskell**, **Groovy** and **Java** for new features and integrations, while **mentoring** junior hires by assigning tasks, guiding and onboarding them on system design; migrated core logic from JavaScript to Haskell, **reducing costs** by 95%
- Designed and maintained a **microservice** based distributed system on **AWS**-EKS (**Kubernetes**) to orchestrate **Docker** containers, RDS and ElastiCache for storage and cache, to process **100M+ daily transactions** using fault-tolerance patterns and load balancing strategies to ensure 99.999% uptime.

Publications

A. N. Krishna, Y. L. Chaitra, Atul M. Bharadwaj, K. T. Abbas, **Allen Abraham**, Anirudh S. Prasad, *"Real-time Machine Vision System for the Visually Impaired"*, Springer Nature, (DOI: 10.1007/s42979-024-02741-4)

Projects

Pegasus - AI-Agentic Job Screening Pipeline

- Architected an **event-driven AI** screening pipeline utilizing **LLMs** (DeepSeek/Qwen) via **OpenRouter** to automate high-accuracy job description analysis and resume matching, generating **structured evaluation metrics**.
- Designed a scalable **publisher-subscriber backend** using a **Python** scraper producer, Local **PII removal**, **Kafka** queues, **PostgreSQL**, and **Docker**; optimized processing using a ThreadPoolExecutor concurrency pool and token-bucket rate limiting

Save For Later

- **Created** and **maintained** an **open-source** Chrome extension (Save for Later), reaching **15,000+ total installations** and sustaining a 4.8-star user rating

eYantra Robotics Competition (by Indian Institute of Technology, Bombay) | *Winner and Team Leader* — 04/**2020**

- **Led** our robotics team to **1st place** among 1,050 teams in a national competition, integrating real-time systems, maze traversal algorithms, image recognition, robotics design, electronics, and machine learning.

Skills

Languages: *Python, C++, Go, Haskell, Java, Groovy, JavaScript*
Technologies: *Git, Linux, SQL, Redis, REST API, TensorFlow, AWS, Docker, Microservices, Claude Code, Kafka, OpenRouter, Deepseek*
Software Development: *Agile methodology, AI driven coding,Scalable System Design, Object-Oriented Programming, LLM Orchestration, Context Engineering, Event-Driven Architecture*